Supporting Dementia Care Through Synthetic Routine Generation and Wearable Anomaly Detection

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Introduction

Dementia currently affects over 55 million people worldwide, a number expected to double by 2050. Monitoring changes in daily routines plays a crucial role in the early detection of behavioral and cognitive decline [1].

This project introduces an intelligent monitoring system that recognizes activities such as mobility, sleep, hygiene, meals, and falls, while detecting deviations from normal behavioral routines.

As illustrated in Figure 1, the system integrates data from both a wrist-worn and an in-ear wearable device [2], enabling continuous, non-invasive, and real-time monitoring of daily behaviors in elderly individuals with dementia [3].

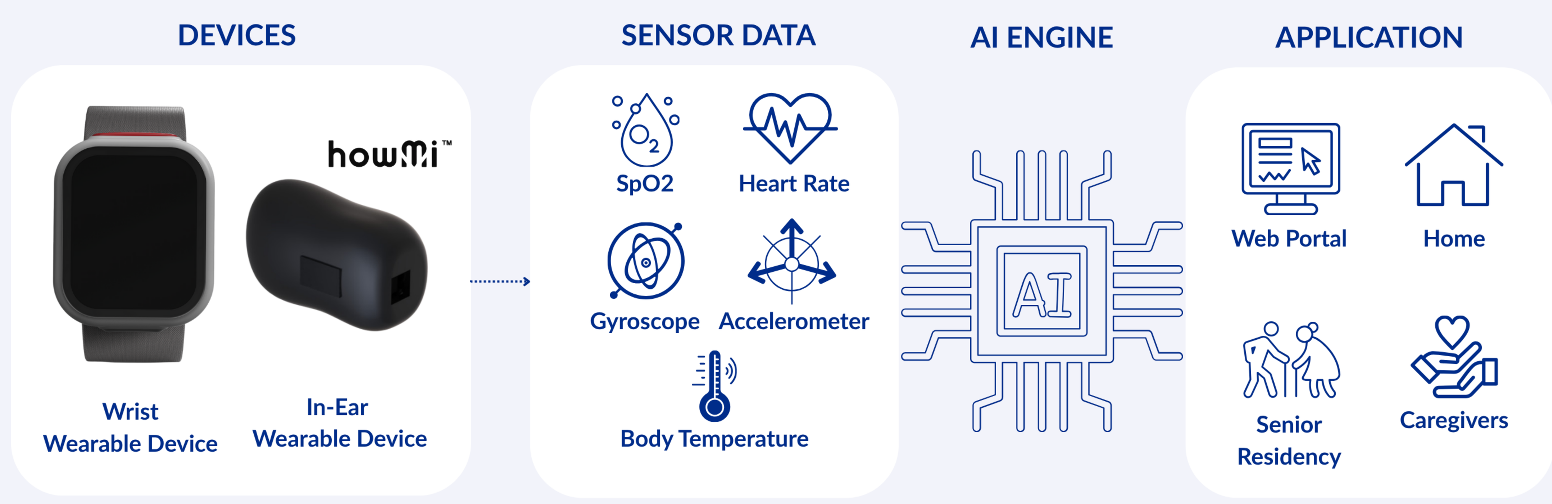

Fig. 1 - System architecture with wearable wrist and ear devices, a local AI engine, and a web-based caregiver portal.

Methodology

1. ROUTINE DESIGN

Synthetic daily routines were generated to simulate realistic 24-hour activity patterns for elderly individuals with dementia. Each day was divided into 288 five-minute intervals, reflecting typical daily structure, and included activities such as personal hygiene, meals, rest, sleep, mobility, and sedentary behavior.

Fig. 2 - Categorization of daily activities used for routine generation, including hygiene, meals, mobility, sedentary behavior, and sleep.

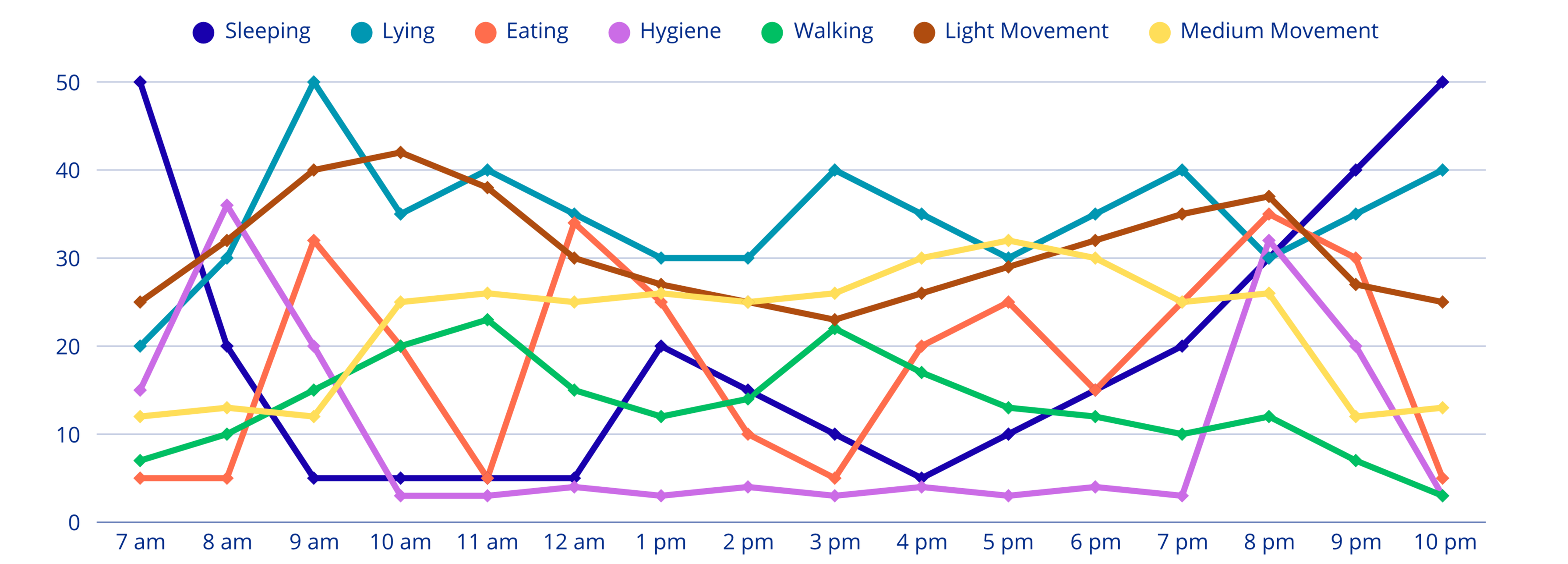

Fig. 3 - Everyday activities during the course of the day. The percentages refer to the proportion of observed activities at the respective time point.

2. ROUTINE VARIABILITY AND ANOMALIES

To capture natural variation, random shifts in timing, duration, and order were introduced. Controlled anomalies (e.g., missed meals or night wandering) were then injected to simulate atypical behaviors observed in dementia.

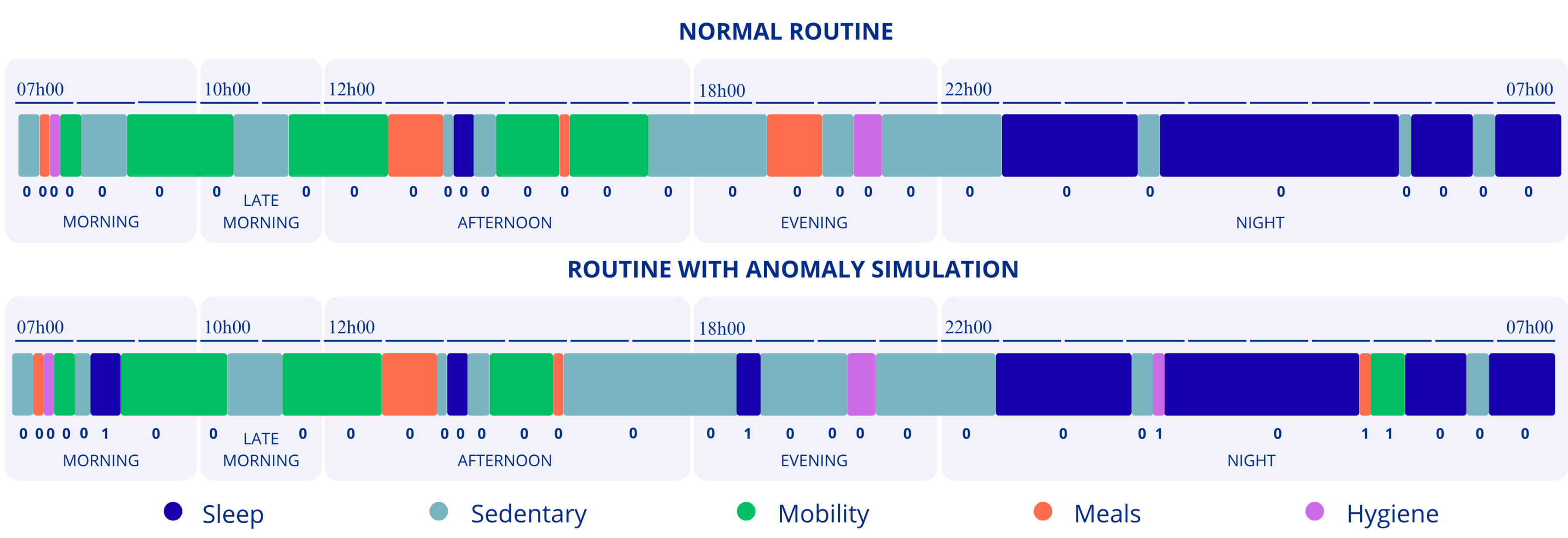

Fig. 4 - Comparison between a normal routine (top), a routine with injected anomalies (bottom), and the corresponding binary anomaly labels.

3. MODEL TRAINING

To evaluate the effectiveness of anomaly detection techniques on the generated dataset, we implemented a sequence model based on Gated Recurrent Units (GRUs) capable of learning temporal dependencies and identifying behavioral deviations.

Results

The GRU-based anomaly detection model was evaluated on a synthetic dataset of 20,000 daily sequences of 288 five-minute intervals. Anomalies were injected into 10,000 sequences, affecting 15% of steps per day. Data were split into 70% training, 15% validation, and 15% testing.

The model achieved an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.999, and an Area Under the Precision–Recall Curve (PR-AUC) of 0.989, and overall accuracy of 99.4%.

	PRECISION	RECALL	F1-SCORE
NORMAL	0.995	0.998	0.997
ANOMALOUS	0.978	0.941	0.959

Fig. 5 - Performance metrics for normal and anomalous classes

Conclusion and Future Work

This work presents a reproducible approach for generating synthetic routines and injecting clinically relevant anomalies, enabling robust training of anomaly detection models. A GRU-based model achieved high accuracy, demonstrating the value of synthetic data in addressing data scarcity and ethical limitations. Future work will focus on validating with real-world wearable data, accounting for variability, and refining anomalies with clinical input.

Acknowledgments

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